Proving the Security of PeerDAS without the AGM

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Abstract

Data availability sampling (DAS) enables clients to verify availability of data without downloading it entirely. This concept is crucial to Ethereum's roadmap. An instantiation of this concept, known as PeerDAS, relies at its core on a variant of KZG polynomial commitments and is set to be integrated into Ethereum. To assess the security of PeerDAS, Wagner and Zapico (ePrint 2024) provided a formal analysis, proving its security as a cryptographic primitive. However, their proof relies on the algebraic group model – an idealized framework known to be uninstantiable (Zhandry, CRYPTO 2022).

In this work, we establish the security of PeerDAS in the standard model under falsifiable assumptions. Specifically, we eliminate reliance on the algebraic group model and instead base our proof on the ARSDH assumption (Lipmaa et al., EUROCRYPT 2024), thus strengthening the theoretical foundations of PeerDAS and enhancing confidence in its security.

Contents

1 Introduction

Data availability sampling (DAS) [ASBK21, HASW23] is a concept designed to improve the scalability of blockchains such as Ethereum. The primary goal of DAS is to allow parties to verify the availability of data on a distributed network without the need to download it entirely. Instead, parties only download a succinct commitment and small random portions of the data. In detail, the data is redundantly encoded using an erasure code. The codeword is then committed to, and clients attempt to download individual symbols of the codeword, verifying them against the commitment.

PeerDAS and Its Analysis. One prominent implementation of DAS is called PeerDAS, based on KZG polynomial commitments [KZG10], which is soon to be integrated into Ethereum. Due to its significance for Ethereum, Wagner and Zapico [WZ24] have proven the security of PeerDAS within the *algebraic group model* [FKL18]. Specifically, they follow the framework of [HASW23] and define an *erasure code commitment scheme* that models the cryptographic layer of PeerDAS. They then demonstrate two essential properties for this scheme, which are sufficient [HASW23] to imply the security of the resulting DAS scheme: *position-binding* and *code-binding*. Intuitively, position-binding ensures that one cannot open a single position of the codeword to two contradicting symbols. Code-binding guarantees that any set of valid openings an adversary might propose is consistent with the erasure code – i.e., there exists a codeword that corresponds to that set of openings. Notably, in PeerDAS's security proof only the proof of code-binding relies on the algebraic group model, while Wagner and Zapico show position-binding in the standard model.

The Algebraic Group Model. The algebraic group model (AGM), introduced by Fuchsbauer, Kiltz, and Loss [FKL18], treats all adversaries as algebraic algorithms. Algebraic algorithms generate their outputs by applying group operations to their inputs. Specifically, when an algorithm outputs a group element Y after having obtained group elements $X_1, \ldots, X_\ell$, it also outputs coefficients $\chi_1, \ldots, \chi_\ell$ such that $Y = \prod_i X_i^{\chi_i}$. While the AGM has been used in many previous works ([GWC19, CHM+20, NRS21, ZGK+22, BL22, BLSW24], just to mention a few), some studies have identified issues with it [Zha22, LPS23, ZZK22]. For example, Zhandry [Zha22] demonstrates that the AGM is un-instantiable by providing a construction of a MAC that is secure in the AGM but insecure in any standard-model instantiation of the group. Lipmaa et al. [LPS23] present a knowledge assumption that holds in the AGM but fails in any realistic group. Katz, Zhang, and Zhou [ZZK22] criticize the AGM for not aligning with the intuition of what operations an algebraic algorithm should be allowed to perform, especially highlighting that the AGM is not strictly weaker than the generic group model (GGM) [Sho97], contrary to previous beliefs.

Our Goal. Given the issues associated with the plain AGM, it has been left as an open problem in [WZ24] to eliminate reliance on the algebraic group model by proving code-binding under weaker idealizations [Lip22, LPS23] or even without idealizations, based solely on falsifiable assumptions. This brings us to the central question of our work:

Can we base the security of Ethereum's PeerDAS *solely on falsifiable assumptions without relying on algebraic group models?*

The simple answer to this question is *yes*, as code-binding itself is already a falsifiable assumption. However, proving security by just assuming security is far from satisfiable. Instead, we want to use assumptions that are already known.

1.1 Overview of Our Result

We improve upon the analysis from [WZ24] and show that code-binding holds for the PeerDAS commitment scheme based on the (falsifiable) ARSDH assumption[1], introduced by [LPS24]. Combining our new result with the findings in [WZ24], we obtain the following informal statement:

Theorem 1 (Main Theorem, Informal). *Under the falsifiable assumptions* BSDH *and* ARSDH*,* PeerDAS *is a secure data availability sampling scheme, as defined in [HASW23].*

[1] We also use position-binding, which was already shown in [WZ24] based on the falsifiable BSDH assumption, without relying on the algebraic group model.

To establish this result, our starting point is the result by Lipmaa, Parisella, and Siim [LPS24]. They show that KZG commitments [KZG10] satisfy a notion of *special soundness* based on the ARSDH assumption[2]. To achieve this, they employ a *batching lemma* for KZG openings, first shown by Tomescu et al. [TAB+20]. Our first observation is that this notion of special soundness, combined with position-binding, implies code-binding when KZG is treated as an erasure code commitment scheme for the Reed-Solomon code. Still, PeerDAS does not use KZG *as is*, but rather a variant of KZG that employs *multiproofs*, a feature not analyzed by Lipmaa et al. We use a generalized batching lemma for KZG openings of [AFZ] to overcome this challenge.

1.2 Additional Related Work

In this section, we discuss prior research relevant to our work. Since our core contribution is establishing a new security property of KZG commitments [KZG10], we highlight related results in this area. Additionally, we discuss results on data availability sampling (DAS).

Work on the Security of KZG. The result by Lipmaa, Parisella, and Siim [LPS24] shows that the KZG polynomial commitment scheme is black-box extractable under a new, falsifiable assumption called ARSDH. This result suggests the possibility that many zkSNARKs can be proven knowledge-sound without relying on the GGM or AGM. They demonstrate this is the case for a (less efficient variant) of PLONK [GWC19]. In a follow-up work [LPS24], the same authors have extended their results to various batching protocols for KZG commitments, deriving in proofs of knowledge-soundness for more efficient variants of the PLONK protocol. Most recently, Belohorec et al. [BDH+25] have analyzed further variants of KZG-based commitment schemes, including the multivariate version. They study knowledge-soundness across a range of protocols that feature “KZG-like” verification checks and introduce a framework to prove it without the AGM.

Work on DAS. Data availability sampling (DAS) was initially introduced by Al-Bassam et al. [ASBK21] and later rigorously formalized as a cryptographic primitive by Hall-Andersen, Simkin, and Wagner [HASW23]. Their work also introduced a general framework for constructing DAS schemes using erasure codes and associated erasure code commitment schemes, capturing nearly all known DAS constructions. Within this framework, Wagner and Zapico [WZ24] provided a security analysis of PeerDAS in the algebraic group model. Hall-Andersen, Simkin, and Wagner [HASW24] explored the connection between interactive oracle proofs of proximity (IOPPs) [BBHR18] and erasure code commitments, leading to a DAS construction based on the FRI IOPP of Ben-Sasson et al. [BBHR18]. More recently, Evans, Mohnblatt, and Angeris [EMA25] aimed to reduce the overhead of such constructions, although their work does not explicitly introduce a new DAS scheme.

2 Preliminaries

Here, we recall relevant definitions and notation.

2.1 Miscellaneous

Notation. We use standard cryptographic terminology (a security parameter λ, PPT, or negligible functions). For an algorithm A, the running time of A is denoted by $\mathbf{T}(\mathsf{A})$. If A is deterministic, we write $y := \mathsf{A}(x)$ to denote that we run it on input x and set y to the output. If A is probabilistic, we write $y \leftarrow \mathsf{A}(x)$ if it is run on uniform random coins. To make the coins ρ explicit, we may write $y := \mathsf{A}(x;\rho)$. Writing $y \in \mathsf{A}(x)$ means that y is a possible output of A on input x. We use $x \stackrel{\$}{\leftarrow} X$ to denote the sampling of x uniformly at random from a finite set X. We denote $[r] = \{1, \ldots, r\} \subseteq \mathbb{N}$. Throughout the document, it will be useful to index vectors by set elements. Namely, for a set S and a set X, we write X^S to denote the set of all sequences $(x_s)_{s\in S}$ with $x_s \in X$ for all $s \in S$.

Erasure Codes. An erasure code in an ambient space Λ^n is a subset $\mathcal{C} \subseteq \Lambda^n$. That is, $\mathcal{C}$ contains codewords of length n with symbols in Λ. We also represent the code by its encoding function $\mathcal{C}\colon \Gamma^k \to \Lambda^n$ mapping

[2]They then show how this can be used to prove the security of SNARKs in the random oracle model, though this part is not relevant for our work.

messages from Γ^k into the code. Erasure codes with *reception efficiency* $t \leq n$ have the property that given any t symbols of a codeword, one can reconstruct the message.

Interpolation and Vanishing Polynomials. We use the notation $I_S^{\mathbf{y}}(X)$ to denote the interpolation polynomial over a set of interpolation points S with evaluations $\mathbf{y}_s, s \in S$. That is, it is the unique polynomial of degree $|S|-1$ such that $I_S^{\mathbf{y}}(s) = \mathbf{y}_s$ for all $s \in S$. We denote the vanishing polynomial over a domain S by $V_S(X) = \prod_{s \in S} X - s$.

Groups and Assumptions. We work with pairing groups $\mathcal{G}$, which are represented by tuples $\mathcal{G} = (q, \mathbb{G}_1, \mathbb{G}_2, \mathbb{G}_T, e, \mathcal{P}_1, \mathcal{P}_2)$. Here, $\mathbb{G}_1, \mathbb{G}_2$ and $\mathbb{G}_T$ are groups of prime order q with generators $\mathcal{P}_1, \mathcal{P}_2$, and $\mathcal{P}_T$ respectively, and $e \colon \mathbb{G}_1 \times \mathbb{G}_2 \to \mathbb{G}_T$ is a pairing, i.e., an efficiently computable, non-degenerate bilinear map. Non-degenerate means that the point $\mathcal{P}_T = e(\mathcal{P}_1, \mathcal{P}_2)$ generates $\mathbb{G}_T$. We work in the type-3 pairing setting [GPS06], where there is no efficiently computable isomorphism between $\mathbb{G}_1$ and $\mathbb{G}_2$. We denote elements in these groups implicitly: $[a]_\gamma := a\mathcal{P}_\gamma$ for $\gamma \in \{1, 2, T\}$. Note that this means that $e([a]_1, [b]_2) = [ab]_T$. We assume that there is some algorithm PGGen that takes as input the security parameter 1^λ and outputs the description of $\mathcal{G}$. Next, we define the two computational hardness assumptions that we rely on. Both assumptions are falsifiable. Namely, we rely on the BSDH assumption, as already used in the previous analysis of PeerDAS [WZ24], and on the ARSDH assumption[3], as introduced by [LPS24].

Definition 1 (BSDH Assumption)**.** Let $d \geq D$. We say that the (d, D)-BSDH assumption holds relative to PGGen, if for every PPT algorithm $\mathcal{A}$ the following advantage is negligible:

$$\mathsf{Adv}^{(d,D)\text{-}\mathsf{BSDH}}_{\mathcal{A},\mathsf{PGGen}}(\lambda) := \Pr\left[c \neq \tau \wedge W = [1/(\tau - c)]_T \;\middle|\; \begin{array}{l} \mathcal{G} \leftarrow \mathsf{PGGen}(1^\lambda),\ \tau \xleftarrow{\$} \mathbb{Z}_q, \\ (c, W) \leftarrow \mathcal{A}(\mathcal{G}, ([\tau^i]_1)_{i=0}^{d}, ([\tau^i]_2)_{i=0}^{D}) \end{array} \right].$$

Definition 2 (ARSDH Assumption)**.** Let $d \geq D$. We say that the (d, D)-ARSDH assumption holds relative to PGGen, if for every PPT algorithm $\mathcal{A}$ the following advantage is negligible:

$$\mathsf{Adv}^{(d,D)\text{-}\mathsf{ARSDH}}_{\mathcal{A},\mathsf{PGGen}}(\lambda) := \Pr\left[\begin{array}{l} S \subseteq \mathbb{Z}_q \wedge |S| = d+1 \wedge [g]_1 \neq [0]_1 \\ \wedge\, e([g]_1, [1]_2) = e([\varphi]_1, [V_S(\tau)]_2) \end{array} \;\middle|\; \begin{array}{l} \mathcal{G} \leftarrow \mathsf{PGGen}(1^\lambda),\ \tau \xleftarrow{\$} \mathbb{Z}_q, \\ (S, [g]_1, [\varphi]_1) \\ \qquad \leftarrow \mathcal{A}(\mathcal{G}, ([\tau^i]_1)_{i=0}^{d}, ([\tau^i]_2)_{i=0}^{D}) \end{array} \right].$$

2.2 KZG Commitments and Multiproofs

PeerDAS is based on the polynomial commitment scheme by Kate, Zaverucha and Goldberg [KZG10]. Given a degree bound $d \in \mathbb{N}$, the KZG commitment scheme enables a party to commit succinctly to a polynomial $f \in \mathbb{Z}_q[X]$ of degree d. Later, this party can compute succinct opening proofs that convince a verifier (holding $x, y \in \mathbb{Z}_q$ and the commitment) that $f(x) = y$. PeerDAS does not use the standard KZG opening procedure. Instead, D evaluations are opened at once in a succinct way, using KZG multiproofs. These groups of D elements are also referred to as *cells* in the context of PeerDAS. We define the relevant KZG algorithms next:

- $\mathsf{KZG.Setup}(1^\lambda, 1^d, 1^D) \to \mathsf{ck}$:
  1. Generate $\mathcal{G} \leftarrow \mathsf{PGGen}(1^\lambda)$, where $\mathcal{G} = (q, \mathbb{G}_1, \mathbb{G}_2, \mathbb{G}_T, e, \mathcal{P}_1, \mathcal{P}_2)$.
  2. Sample $\tau \xleftarrow{\$} \mathbb{Z}_q$ and set commitment key $\mathsf{ck} := \big(\mathcal{G}, d, ([\tau^i]_1)_{i=0}^{d}, ([\tau^i]_2)_{i=0}^{D}\big)$.
- $\mathsf{KZG.Com}(\mathsf{ck}, f) \to \mathsf{com}$ for $f \in \mathbb{Z}_q[X]$ with $\deg f \leq d$:
  1. Write $f(X) = \sum_{i=0}^{d} f_i X^i$.
  2. Set $\mathsf{com} = [c]_1 = [f(\tau)]_1 = \sum_{i=0}^{d} f_i [\tau^i]_1$.

[3]In Lipmaa et al.'s version, the adversary gets only $[1]_2$ and $[\tau]_2$ in the second source group, whereas we need a slightly stronger version in which the adversary gets $([\tau^i]_2)_{i=0}^{D}$. This is due to the use of multiproofs. We are confident that this does not make the assumption significantly stronger. In particular, the assumption has been introduced by Limpaa et al. as an adaptive variant of the RSDH assumption (from [GR19]), in which the adversary gets even more powers in the second source group. Further, the heuristic hardness analysis of Lipmaa et al. can trivially be extended to our version.

- $\mathsf{KZG.MultiOpen}(\mathsf{ck}, f, S) \to \pi$ for $f \in \mathbb{Z}_q[X]$ with $\deg f \leq d$ and $S \subset \mathbb{Z}_q$, $|S| \leq D$:
  1. Set $\mathbf{y} \in \mathbb{Z}_q^S$ with $\mathbf{y}_s := f(s)$ for all $s \in S$.
  2. Let $Q(X) := (f(X) - I_S^{\mathbf{y}}(X))/V_S(X)$ and observe that $Q \in \mathbb{Z}_q[X]$ and $\deg Q \leq d$.
  3. Write $Q(X) = \sum_{i=0}^{d} Q_i X^i$ and set $\pi := [\varphi]_1 := [Q(\tau)]_1 = \sum_{i=0}^{d} Q_i [\tau^i]_1$.
- $\mathsf{KZG.MultiVer}(\mathsf{ck}, \mathsf{com}, S, \mathbf{y}, \pi) \to b$ for $S \subset \mathbb{Z}_q$ with $|S| \leq D$, $\mathbf{y} \in \mathbb{Z}_q^S$, and $\mathsf{com}, \pi \in \mathbb{G}_1$:
  1. Write $\mathsf{com} = [c]_1$ and $\pi = [\varphi]_1$.
  2. Compute $V_S(X)$ and from that $[V_S(\tau)]_2$ using $([\tau^i]_2)_{i=0}^{D}$.
  3. If $e([c]_1 - [I(\tau)]_1, [1]_2) = e([\varphi]_1, [V_S(\tau)]_2)$, return $b := 1$. Otherwise, return $b := 0$.

This scheme is correct, i.e., honestly committing and opening makes the verification algorithm accept.

2.3 Generalized Batching Lemma

In [LPS24], the authors used a batching lemma by Tomescu et al. [TAB+20] to show special soundness of KZG (without multiproofs) from ARSDH. The batching lemma states that, given KZG proofs of openings $\{\pi_s\}_{s \in S}$ attesting $f(s) = \mathbf{y}_s$, it is possible to aggregate them into a single multiproof π_S, equal to the output of the KZG.MultiOpen algorithm described in Section 2.2. In [AFZ] the authors generalized this result: given KZG multiproofs of openings $\{\pi_{S_j}\}_{j \in J}$ it is possible to aggregate them into a single proof π_{I_S} that attests $f(s) = \mathbf{y}_s$ for all $s \in S_j, j \in J$. Notably, the result holds only when all S_j correspond to coset of a subgroup of a group of roots of unity, as it is the case in PeerDAS. We present the result from [AFZ] below.

Let $\mathbb{H} \subset \mathbb{Z}_q$ be a group of roots of unity, i.e., $\mathbb{H} = \{1, \omega, \omega^2, \ldots, \omega^{m-1}\}$ for an mth root of unity ω. We fix D that divides m (below, we will have $m = nD$) and a subgroup $\mathbb{S} = \{1, \nu, \nu^2, \ldots, \nu^D\} \subseteq \mathbb{H}$ of size D, generated by $\nu \in \mathbb{Z}_q$. A *coset* of $\mathbb{S}$ in $\mathbb{H}$ is a set of the form $\omega^j \mathbb{S}$. For completeness, we state below the Multiproof Batching Lemma from [AFZ] as well as the intermediate results that prove it.

Lemma 1. *Let $S_i = \omega^i \mathbb{S}$ and $S_j = \omega^j \mathbb{S}$ be cosets of $\mathbb{S}$ in $\mathbb{H}$. Then, for every $s \in S_i$, we have*

$$V_{S_j}(s) = \omega^{iD} - \omega^{jD}.$$

In particular, we have $V_{S_j}(s) = V_{S_j}(s')$ for all $s, s' \in S_i$.

Proof. First, one can see that the vanishing polynomials have the form $V_{S_j}(X) = X^D - \omega^{jD}$. To see this, note that this polynomial vanishes on all $s \in S_j$ and has the right degree:

$$\text{For } s = \omega^j \nu^c \in S_j : \; : \; s^D - \omega^{jD} = \omega^{jD} \nu^{cD} - \omega^{jD} = \omega^{jD} - \omega^{jD} = 0,$$

where we have used that ν has order D. Now, let $s = \omega^i \nu^r \in S_i$. Then, we have

$$V_{S_j}(s) = s^D - \omega^{jD} = \omega^{iD} \nu^{rD} - \omega^{jD} = \omega^{iD} - \omega^{jD}.$$

□

Lemma 2 (Multiproof Batching Lemma)**.** *Let $S \subseteq \mathbb{Z}_q$ be a set and consider a partitioning $S = S_1 \dot\cup \cdots \dot\cup S_\ell$ of S into disjoint sets S_i, where we assume that each S_i is a coset of $\mathbb{S}$ in $\mathbb{H}$. Consider field elements $c, \tau \in \mathbb{Z}_q$ and $\varphi_j \in \mathbb{Z}_q$ for every $j \in [\ell]$. Consider a vector $\mathbf{y} \in \mathbb{Z}_q^S$ with entries $\mathbf{y}_s \in \mathbb{Z}_q$ for every $s \in S$, and for each $j \in [\ell]$, write $\mathbf{y}^{(j)} \in \mathbb{Z}_q^{S_j}$ to denote the part of $\mathbf{y}$ related to $S_j \subseteq S$. Then, under the assumption that*

$$\forall j \in [\ell] \colon \; c - I_{S_j}^{\mathbf{y}^{(j)}}(\tau) = \varphi_j \cdot V_{S_j}(\tau),$$

we have that

$$c - I_S^{\mathbf{y}}(\tau) = \varphi \cdot V_S(\tau),$$

for $\varphi = \sum_{j \in [\ell]} \Delta_j \varphi_j$ where $\Delta_j = 1/V_{S \setminus S_j}(s)$ for any $s \in S_j$.

Proof. First, note that the value of Δ_j is well-defined, due to Lemma 1. We start the proof with some claims. Before we give them, we introduce notation regarding Lagrange polynomials. Namely, we define

$$\mu_s(X) := \prod_{s' \in S, s' \neq s} \frac{X - s'}{s - s'} \quad \text{and} \quad \mu_s^j(X) := \prod_{s' \in S_j, s' \neq s} \frac{X - s'}{s - s'} \quad \text{for all } j \in [\ell].$$

With that, first recall that

$$I_S^{\mathbf{y}}(X) = \sum_{s \in S} \mathbf{y}_s \mu_s(X) \quad \text{and} \quad I_{S_j}^{\mathbf{y}^{(j)}}(X) = \sum_{s \in S_j} \mathbf{y}_s \mu_s^j(X) \quad \text{for all } j \in [\ell].$$

Claim 1. We have $I_S^{\mathbf{y}}(X) = \sum_{j \in [\ell]} \Delta_j V_{S \setminus S_j}(X) I_{S_j}^{\mathbf{y}^{(j)}}(X)$.

Proof of Claim. We have

$$\begin{aligned}
I_S^{\mathbf{y}}(X) &= \sum_{s \in S} \mathbf{y}_s \mu_s(X) = \sum_{j \in [\ell]} \sum_{s \in S_j} \mathbf{y}_s \mu_s(X) = \sum_{j \in [\ell]} \sum_{s \in S_j} \mathbf{y}_s \prod_{i \in S, i \neq s} \frac{X - i}{s - i} \\
&= \sum_{j \in [\ell]} \sum_{s \in S_j} \mathbf{y}_s \prod_{i \in S_j, i \neq s} \frac{X - i}{s - i} \prod_{i \in S \setminus S_j} \frac{X - i}{s - i} \\
&= \sum_{j \in [\ell]} \sum_{s \in S_j} \mathbf{y}_s \mu_s^j(X) \frac{V_{S \setminus S_j}(X)}{V_{S \setminus S_j}(s)} = \sum_{j \in [\ell]} \sum_{s \in S_j} \Delta_j \mathbf{y}_s \mu_s^j(X) V_{S \setminus S_j}(X) \\
&= \sum_{j \in [\ell]} \Delta_j V_{S \setminus S_j}(X) \sum_{s \in S_j} \mathbf{y}_s \mu_s^j(X) = \sum_{j \in [\ell]} \Delta_j V_{S \setminus S_j}(X) I_{S_j}^{\mathbf{y}^{(j)}}(X).
\end{aligned}$$

Claim 2. We have $1/V_S(X) = \sum_{j \in [\ell]} \Delta_j / V_{S_j}(X)$.

Proof of Claim. We first define the formal derivative $V_S'(X)$ of $V_S(X)$, which is

$$V_S'(X) = \sum_{s \in S} \frac{V_S(X)}{(X - s)} \in \mathbb{Z}_q[X], \text{ and thus, } V_S'(s) = \prod_{i \in S, i \neq s} (s - i) \text{ for all } s \in S.$$

Now, using this definition, we get

$$\begin{aligned}
\frac{1}{V_S(X)} &= \sum_{s \in S} \frac{1}{V_S'(s)} \frac{1}{(X - s)} = \sum_{j \in [\ell]} \sum_{s \in S_j} \frac{1}{V_S'(s)} \frac{1}{(X - s)} \\
&= \sum_{j \in [\ell]} \sum_{s \in S_j} \frac{1}{\prod_{i \in S, i \neq s}(s - i)} \frac{1}{(X - s)} = \sum_{j \in [\ell]} \sum_{s \in S_j} \frac{1}{V_{S \setminus S_j}(s)} \frac{1}{V_{S_j}'(s)(X - s)} \\
&= \sum_{j \in [\ell]} \Delta_j \sum_{s \in S_j} \frac{1}{V_{S_j}'(s)(X - s)} = \sum_{j \in [\ell]} \Delta_j \frac{1}{V_{S_j}(X)}.
\end{aligned}$$

Completing the Proof. Now, we use these two claims to complete the proof. We want to show that $c - I_S^{\mathbf{y}}(\tau) = \varphi \cdot V_S(\tau)$. To this end, we start with $(c - I_S^{\mathbf{y}}(\tau))/V_S(\tau)$ and show that it equals φ, using our two claims. Namely, using our claims first and then that $c - I_{S_j}^{\mathbf{y}^{(j)}}(\tau) = \varphi_j \cdot V_{S_j}(\tau)$ for all j, we get

$$\begin{aligned}
\frac{c - I_S^{\mathbf{y}}(\tau)}{V_S(\tau)} &= \frac{c}{V_S(\tau)} - \frac{I_S^{\mathbf{y}}(\tau)}{V_S(\tau)} \\
&= c \sum_{j \in [\ell]} \Delta_j \frac{1}{V_{S_j}(\tau)} - \frac{\sum_{j \in [\ell]} \Delta_j V_{S \setminus S_j}(\tau) I_{S_j}^{\mathbf{y}^{(j)}}(\tau)}{V_S(\tau)} \\
&= \sum_{j \in [\ell]} \Delta_j \left(\frac{c}{V_{S_j}(\tau)} - \frac{V_{S \setminus S_j}(\tau) I_{S_j}^{\mathbf{y}^{(j)}}(\tau)}{V_S(\tau)} \right) \\
&= \sum_{j \in [\ell]} \Delta_j \left(\frac{c}{V_{S_j}(\tau)} - \frac{I_{S_j}^{\mathbf{y}^{(j)}}(\tau)}{V_{S_j}(\tau)} \right) = \sum_{j \in [\ell]} \Delta_j \frac{c - I_{S_j}^{\mathbf{y}^{(j)}}(\tau)}{V_{S_j}(\tau)} = \sum_{j \in [\ell]} \Delta_j \varphi_j = \varphi.
\end{aligned}$$

$\square$

2.4 Erasure Code Commitments

To modularize the analysis of data availability sampling (DAS), Hall-Andersen, Simkin, and Wagner [HASW23] have introduced the notion of *erasure code commitments*. Such erasure code commitments model the cryptographic core of most DAS schemes and can be compiled to DAS via their framework. PeerDAS is also captured by this framework, which means that in this document we can focus on the associated erasure code commitment scheme. Intuitively, an erasure code commitment scheme allows to commit to a codeword of an erasure code and later open individual positions of the codeword. The scheme should be *position-binding*, i.e., one cannot open one position to two different symbols, and *code-binding*, i.e., any set of openings is consistent with a codeword. The latter is what differentiates erasure code commitments from regular vector commitments over codewords.

Definition 3 (Erasure Code Commitment Scheme)**.** Let $\mathcal{C}\colon \Gamma^k \to \Lambda^n$ be an erasure code. An erasure code commitment scheme for $\mathcal{C}$ is a tuple $\mathsf{CC} = (\mathsf{CC.Setup}, \mathsf{CC.Com}, \mathsf{CC.Open}, \mathsf{CC.Ver})$ of PPT algorithms, with the following syntax:

- $\mathsf{CC.Setup}(1^\lambda) \to \mathsf{ck}$: takes as input the security parameter and outputs a commitment key ck.
- $\mathsf{CC.Com}(\mathsf{ck}, m) \to (\mathsf{com}, St)$: takes as input a commitment key ck and a string $m \in \Gamma^k$, and outputs a commitment com and a state St.
- $\mathsf{CC.Open}(\mathsf{ck}, St, j) \to \pi$: takes as input a commitment key ck, a state St, and an index $j \in [n]$, and outputs an opening π.
- $\mathsf{CC.Ver}(\mathsf{ck}, \mathsf{com}, j, \hat{m}_j, \pi) \to b$: is deterministic, takes as input a commitment key ck, a commitment com, and index $j \in [n]$, a symbol $\hat{m}_j \in \Lambda$, and an opening π, and outputs a bit $b \in \{0,1\}$.

Further, we require that the following completeness property holds: For every $\mathsf{ck} \in \mathsf{Setup}(1^\lambda)$, every $m \in \Gamma^k$, and every $j \in [n]$, we have

$$\Pr\left[\mathsf{CC.Ver}(\mathsf{ck}, \mathsf{com}, j, \hat{m}_j, \pi) = 1 \;\middle|\; \begin{array}{l} (\mathsf{com}, St) \leftarrow \mathsf{CC.Com}(\mathsf{ck}, m), \\ \hat{m} := \mathcal{C}(m),\ \pi \leftarrow \mathsf{CC.Open}(\mathsf{ck}, St, j) \end{array}\right] = 1.$$

Definition 4 (Position-Binding of CC)**.** Let CC be an erasure code commitment scheme for an erasure code $\mathcal{C}$. We say that CC is position-binding if for every PPT algorithm $\mathcal{A}$ the following advantage is negligible:

$$\mathsf{Adv}^{\mathsf{pos\text{-}bind}}_{\mathcal{A},\mathsf{CC}}(\lambda) := \Pr\left[\begin{array}{rl} & \hat{m}_j \neq \hat{m}'_j \\ \wedge & \mathsf{CC.Ver}(\mathsf{ck}, \mathsf{com}, j, \hat{m}_j, \pi) = 1 \\ \wedge & \mathsf{CC.Ver}(\mathsf{ck}, \mathsf{com}, j, \hat{m}'_j, \pi') = 1 \end{array} \;\middle|\; \begin{array}{l} \mathsf{ck} \leftarrow \mathsf{CC.Setup}(1^\lambda), \\ (\mathsf{com}, j, \hat{m}_j, \pi, \hat{m}'_j, \pi') \leftarrow \mathcal{A}(\mathsf{ck}) \end{array}\right].$$

Definition 5 (Code-Binding of CC)**.** Let CC be an erasure code commitment scheme for an erasure code $\mathcal{C}$. We say that CC is code-binding if for every PPT algorithm $\mathcal{A}$ the following advantage is negligible:

$$\mathsf{Adv}^{\mathsf{code\text{-}bind}}_{\mathcal{A},\mathsf{CC}}(\lambda) := \Pr\left[\begin{array}{c} \neg\left(\exists c \in \mathcal{C}(\Gamma^k) : \forall j \in J : c_j = \hat{m}_j\right) \\ \wedge\, \forall j \in J :\ \mathsf{CC.Ver}(\mathsf{ck}, \mathsf{com}, j, \hat{m}_j, \pi_j) = 1 \end{array} \;\middle|\; \begin{array}{l} \mathsf{ck} \leftarrow \mathsf{CC.Setup}(1^\lambda), \\ (\mathsf{com}, J, (\hat{m}_j, \pi_i)_{j \in J}) \leftarrow \mathcal{A}(\mathsf{ck}) \end{array}\right].$$

3 PeerDAS without the AGM

Here, we present our main result, namely, a security proof of PeerDAS without the algebraic group model. We use the generalized batching lemma from previous section to prove code-binding of the *row commitment* of PeerDAS, in the terminology of [WZ24]. This is the only part of the analysis in [WZ24] that relied on the algebraic group model. Finally, we discuss how the rest of the analysis proceeds, essentially summarizing the analysis from [WZ24].

3.1 The Erasure Code Commitment

We now present the erasure code and erasure code commitment scheme that forms the cryptographic core of PeerDAS. For that, we use the same notation $\mathbb{H}, \mathbb{S}$ as in Section 2.3.

Erasure Code. The cosets of $\mathbb{S}$ in $\mathbb{H}$ form a partition of $\mathbb{H}$ into cells C_i of equal size $|C_i| = n$, i.e., $C = C_1 \dot\cup \ldots \dot\cup C_n$, where $C_i = \omega^i \mathbb{S}$. Let $k \leq n$. We consider the following variant of the Reed-Solomon code

$$\mathcal{C} = \left\{ \mathbf{y} = \left(\mathbf{y}^{(1)}, \ldots, \mathbf{y}^{(n)}\right) \in \mathbb{Z}_q^{C_1} \times \cdots \times \mathbb{Z}_q^{C_n} \;\middle|\; \begin{array}{r} \exists f \in \mathbb{Z}_p[X], \deg f < k \leq n \colon \\ \forall i \in [n]\ \forall x \in C_i \colon\ \mathbf{y}_x = f(x) \end{array} \right\}.$$

Naturally, we can view $\mathcal{C}$ as being a subset of $(\mathbb{Z}_q^D)^n$, and the encoding function takes f and outputs the corresponding codeword. Intuitively, this corresponds to a standard Reed-Solomon code of dimension kD and length nD, but grouping together D elements into single symbols. We also refer to these symbols (and the cosets C_i) as *cells*.

Erasure Code Commitment. We consider the following erasure code commitment scheme CC for the code $\mathcal{C}$:

- $\mathsf{CC.Setup}(1^\lambda) \to \mathsf{ck}$
  1. Sample $\mathsf{ck} \leftarrow \mathsf{KZG.Setup}(1^\lambda, 1^d, 1^D)$ for $d = kD - 1$.
- $\mathsf{CC.Com}(\mathsf{ck}, f) \to (\mathsf{com}, St)$ for $f \in \mathbb{Z}_q[X]$ with $\deg f < Dk$
  1. Set $\mathsf{com} := \mathsf{KZG.Com}(\mathsf{ck}, f)$ and $St := f$.
- $\mathsf{CC.Open}(\mathsf{ck}, St, j) \to \pi$ for $j \in [n]$
  1. Parse $St = f$ and set $\pi := \mathsf{KZG.MultiOpen}(\mathsf{ck}, f, C_j)$.
- $\mathsf{CC.Ver}(\mathsf{ck}, \mathsf{com}, j, \mathbf{y}^{(j)}, \pi) \to b$ for $i \in [n]$ and $\mathbf{y}^{(j)} \in \mathbb{Z}_q^{C_j}$
  1. Set $b := \mathsf{KZG.MultiVer}(\mathsf{ck}, \mathsf{com}, C_j, \mathbf{y}^{(j)}, \pi)$.

Remark 1. As the C_j are cosets of a subgroup of roots of unity, vanishing polynomials used in $\mathsf{KZG.MultiVer}$ have a specific form. All of this is made explicit in [WZ24].

The following lemma states position-binding and has been shown in [WZ24].

Lemma 3 ([WZ24], Lemma 2)**.** *Let $d = kD - 1$. If the (d, D)-BSDH assumption holds relative to PGGen, then CC as above is position-binding. Concretely, for any PPT algorithm $\mathcal{A}$, there is a PPT algorithm $\mathcal{B}$ with $\mathbf{T}(\mathcal{B}) \approx \mathbf{T}(\mathcal{A})$ and*

$$\mathsf{Adv}^{\mathsf{pos\text{-}bind}}_{\mathcal{A},\mathsf{CC}}(\lambda) \leq \mathsf{Adv}^{(d,D)\text{-}\mathsf{BSDH}}_{\mathcal{B},\mathsf{PGGen}}(\lambda).$$

We give a new proof for code-binding. Importantly, in contrast to the proof in [WZ24], we do not rely on the algebraic group model. Instead, we prove code-binding in the standard model under the (falsifiable!) ARSDH assumption.

Lemma 4. *Let $d = kD - 1$. If the (d, D)-BSDH assumption holds relative to PGGen and the (d, D)-ARSDH assumption holds relative to PGGen, then CC as above is code-binding. Concretely, for any PPT algorithm $\mathcal{A}$, there is are PPT algorithms $\mathcal{B}_1, \mathcal{B}_2$ with $\mathbf{T}(\mathcal{B}_1) \approx \mathbf{T}(\mathcal{A}) \approx \mathbf{T}(\mathcal{B}_2)$ and*

$$\mathsf{Adv}^{\mathsf{code\text{-}bind}}_{\mathcal{A},\mathsf{CC}}(\lambda) \leq \mathsf{Adv}^{(d,D)\text{-}\mathsf{BSDH}}_{\mathcal{B}_1,\mathsf{PGGen}}(\lambda) + \mathsf{Adv}^{(d,D)\text{-}\mathsf{ARSDH}}_{\mathcal{B}_2,\mathsf{PGGen}}(\lambda).$$

Proof. In the code-binding game, the adversary $\mathcal{A}$ gets as input a commitment key

$$\mathsf{ck} = \left(\mathcal{G}, d, ([\tau^i]_1)_{i=0}^d, ([\tau^i]_2)_{i=0}^D\right).$$

It outputs a commitment $\mathsf{com} = [c]_1 \in \mathbb{G}_1$, a set of positions $J \subseteq [n]$, each pointing to a cell, and an opening for each of those cells. If $\mathcal{A}$ wins, then all openings verify, and there is no polynomial $f \in \mathbb{Z}_q[X]$ of degree at most d that is consistent with these openings. In particular, we must have $|J| > k$, as otherwise, there is always such a polynomial f. For convenience, we can assume that $|J| = k+1$. Also, write $J = \{j_1, \dots, j_{k+1}\}$ and define $S_i := C_{j_i}$ to be the domain of the cell corresponding to the ith opening, $i \in [k+1]$. That is, the ith opening that the adversary provides has the form $(\mathbf{y}^{(i)}, [\varphi_i]_1) \in \mathbb{Z}_q^{S_i} \times \mathbb{G}_1$. Let $S := S_1 \dot\cup \cdots \dot\cup S_k$ and $I^* \in \mathbb{Z}_q[X]$ be the unique polynomial of degree at most d that is consistent with all but the last opening, i.e.,

$$I^*(X) := I_S^{\mathbf{y}}(X) \text{ with } \mathbf{y} \in \mathbb{Z}_q^S \text{ such that } \mathbf{y}_s = \mathbf{y}_s^{(i)} \text{ if } s \in S_i.$$

Because $\mathcal{A}$ breaks code-binding, we know that I^* is inconsistent with the provided openings for S_{k+1}, i.e., there is an $s \in S_{k+1}$ such that $I^*(s) \neq \mathbf{y}_s^{(k+1)}$. Now, we consider three events:

- Win: This event occurs, if $\mathcal{A}$ wins the code-binding game, i.e., all openings verify and there is no polynomial of correct degree consistent with all openings.
- Equal: This event occurs if we have $I^*(\tau) = c$.
- $\mathsf{NotEqual}$: This event occurs if we have $I^*(\tau) \neq c$.

Clearly, we have

$$\mathsf{Adv}_{\mathcal{A},\mathsf{CC}}^{\mathsf{code\text{-}bind}}(\lambda) \leq \Pr[\mathsf{Win}] \leq \Pr[\mathsf{Win} \wedge \mathsf{Equal}] + \Pr[\mathsf{Win} \wedge \mathsf{NotEqual}].$$

We bound both terms separately.

Case 1. $I^*(\tau) = c$. We first want to bound the probability of $\mathsf{Win} \wedge \mathsf{Equal}$. In this case, note that we can build a reduction $\mathcal{B}^*$ that breaks position-binding: we output the commitment $\mathsf{com} = [c]_1$, the opening for S_{k+1} provided by the adversary, i.e., $(\mathbf{y}^{(k+1)}, [\varphi_{k+1}]_1)$, and an opening computed for S_{k+1} computed honestly from the polynomial $I^*(X)$, which verifies by correctness of KZG. We get

$$\Pr[\mathsf{Win} \wedge \mathsf{Equal}] \leq \mathsf{Adv}_{\mathcal{B}^*,\mathsf{CC}}^{\mathsf{pos\text{-}bind}}(\lambda) \leq \mathsf{Adv}_{\mathcal{B}_1,\mathsf{PGGen}}^{(d,D)\text{-}\mathsf{BSDH}}(\lambda),$$

for a reduction $\mathcal{B}_1$ that exists due to Lemma 3.

Case 2. $I^*(\tau) \neq c$. We want to bound the probability of $\mathsf{Win} \wedge \mathsf{NotEqual}$. In this case, we break the ARSDH assumption. This is done using Lemma 2 with $\ell = k$. Namely, the reduction $\mathcal{B}_2$ computes I^* from the provided openings for $S_1, \dots, S_k$. Because these openings verify, the precondition of Lemma 2 holds, and so we get

$$c - I_S^{\mathbf{y}}(\tau) = \varphi \cdot V_S(\tau) \text{ for } \varphi = \sum_{j \in [\ell]} \Delta_j \varphi_j. \tag{1}$$

Note that the reduction can efficiently compute $[\varphi]_1$ from the $[\varphi_j]_1$ provided by $\mathcal{A}$. Note that the reduction can also compute $[I_S^{\mathbf{y}}(\tau)]_1$ efficiently from the commitment key, as the degree of $I_S^{\mathbf{y}}(X)$ is $kD - 1 = d$. It then outputs

$$(S, [g]_1, [\varphi]_1) \quad \text{for} \quad [g]_1 := [c]_1 - [I_S^{\mathbf{y}}(\tau)]_1$$

to the ARSDH game. The reduction breaks ARSDH because of Equation (1) and $c - I^*(\tau) \neq 0$. We get

$$\Pr[\mathsf{Win} \wedge \mathsf{NotEqual}] \leq \mathsf{Adv}_{\mathcal{B}_2,\mathsf{PGGen}}^{(d,D)\text{-}\mathsf{ARSDH}}(\lambda).$$

□

3.2 Completing the Picture

It remains to explain how the security of CC, as established in Section 3.1, relates to the security of PeerDAS. This discussion primarily summarizes the work of [WZ24], to which we refer the reader for further details.

In PeerDAS, data is structured into ℓ so-called *blobs*, where each blob consists of kD field elements. Conceptually, these blobs form the rows of a $\ell \times kD$ matrix of field elements representing the data. Alternatively, since each cell contains D field elements, the data can also be viewed as a $\ell \times k$ matrix of cells. The commitment scheme CC, whose security we have analyzed, is then applied independently to each blob (i.e., row), producing ℓ commitments, corresponding codewords (referred to as *extended blobs*), and openings for all cells. Consequently, we obtain a matrix of dimensions $\ell \times n$, where each entry represents a cell containing D field elements. Cells within the same column are grouped into symbols, which are then downloaded (along with their openings) by clients during data availability sampling. As demonstrated in [WZ24, Section 3.2], the security properties of this new commitment scheme – namely, position-binding and code-binding – directly follow from the security of the row-wise commitment CC *in the standard model*. Subsequently, the framework of [HASW23] is applied to get a secure data availability sampling scheme, which constitutes PeerDAS. Since we have proven the security of CC in the standard model, it follows that PeerDAS is secure in the standard model as well.

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